Kochi startup builds India's first commercial underwater drone

India's first commercial remotely-operated vehicle (ROV)/underwater drone, named EyeROV TUNA, has been developed by EyeROV Technologies, a Kochi-based startup.

EyeROV TUNA is a smart micro-ROV, with operational capabilities in harsh and mission-critical underwater environments. It has been designed to perform visual inspection and surveys of submerged structures as a cost-effective underwater rover that works up to a depth of hundred metres. It can be e-controlled using a laptop or joystick. A camera fitted on the underwater drone gives a live video feed of the submarine environment. The product is equipped to perform a variety of functions, including inspection of ship hulls, fish farms, dams, port structures and bridge foundations.

The underwater robotic drone has been handed over to Naval Physical and Oceanographic Laboratory (NPOL) of Defence Research and Development Organisation (DRDO).

The robotic drone, EyeROV TUNA, will send real-time videos of ships to help with their repair and maintenance (Credit: www.eyerov.com)

IIIT Hyderabad and Intel trying to help drivers avoid potholes

Intel, the US chipmaker, is working on a software that will alert drivers about road conditions, including potholes and the surrounding environment, well in advance. The company has created a database of roads in Hyderabad and Bengaluru, and believes that the rate of accidents can be brought down through a proper understanding of the roads.

Intel, with help from International Institute of Information Technology (IIIT), Hyderabad, has created a dataset that detects people, vehicles, animals, vegetation, immovable structures and traffic signs on the road. The company is also in talks with startups and academic researchers who are capable of using the dataset to build a technology that can help drivers react to a variety of potential hazards in a split-second.

According to Nivruti Rai, country head, Intel, about seventeen people die in road accidents per hour; most of them are young men. Artificial intelligence (AI) and vision technologies can be leveraged to give drivers a few extra seconds to avoid collisions.

Alphabet is using AI to create robots that can learn on their own

Alphabet's robot sorting trash (Credit: The Verge)

Alphabet's X Moonshot division, formerly known as Google X, has unveiled its Everyday Robot project, whose goal is to develop a general-purpose learning robot. The idea is to equip the robot with cameras and complex machine learning software, letting it observe the world around and learn from it without needing to be taught every potential situation it may encounter.

The project team is testing robots that can help out in workplace environments. Currently, these early robots are focused on learning how to sort trash. It might sound mundane, but it is tough to get robots to identify different types of objects and then to grasp them. Alphabet X claims that its robots are currently putting less than five per cent of trash in the wrong place, versus an error rate of twenty per cent among the office's humans.